



REPORT

Khayelitsha Community Library Management System



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Introduction

The Khayelitsha Community Library Management System is a C# Windows Forms application connected to a SQL Server relational database. The system is designed to manage library members, book titles, physical book copies, staff and loans. It also displays CRUD functionality, loan and return processing, searching, filtering and reporting.

System Design

The system uses Member, BookTitle, BookCopy, Staff and Loan entities. Primary keys uniquely identify records and foreign keys maintain relationships between related tables.

Relational Schema

Member: MemberID (PK), FullName, Address, Phone, JoinDate

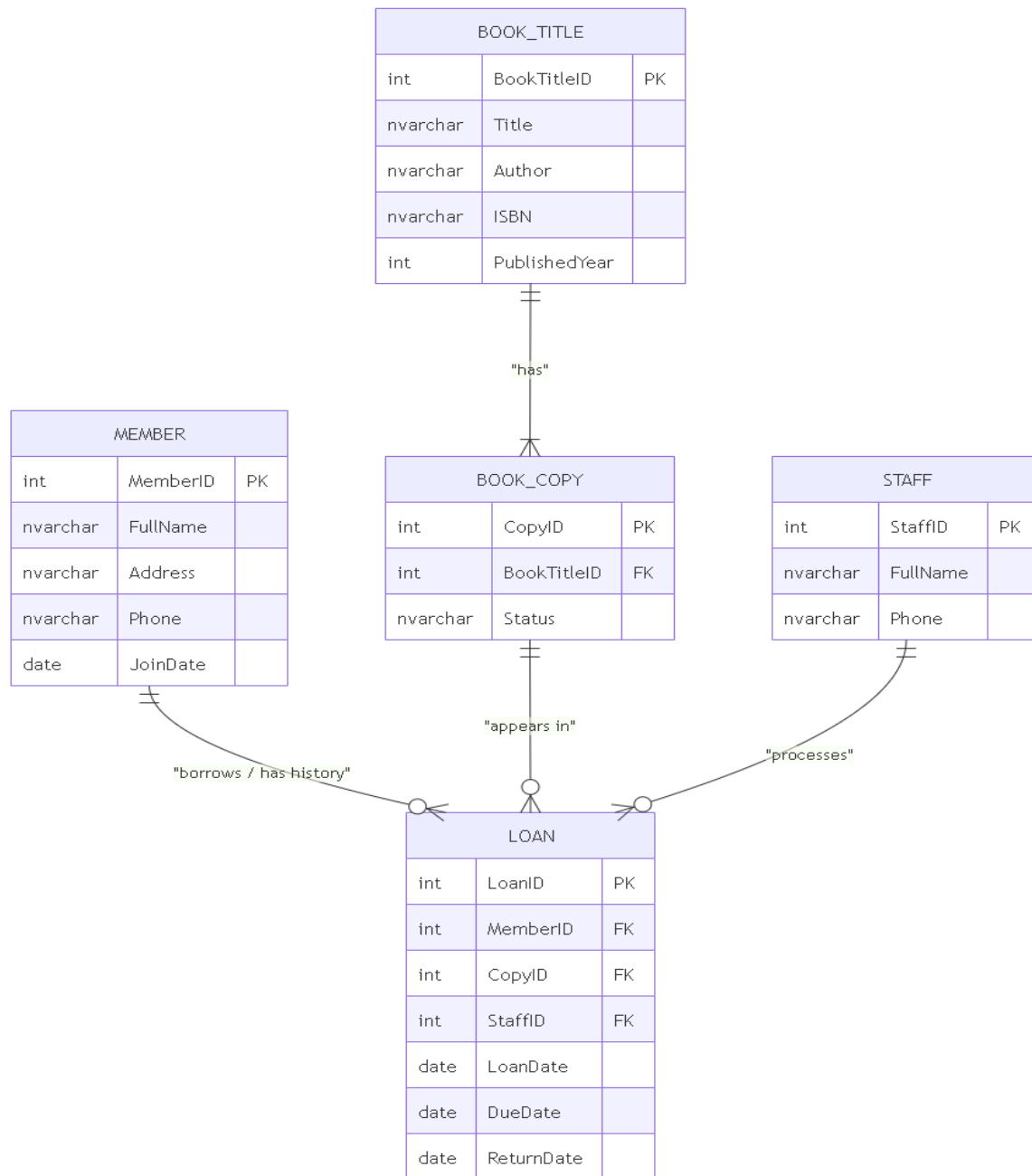
BookTitle: BookTitleID (PK), Title, Author, ISBN, PublicationYear

BookCopy: CopyID (PK), BookTitleID (FK), Status

Staff: StaffID (PK), FullName

Loan: LoanID (PK), MemberID (FK), CopyID (FK), StaffID (FK), LoanDate, DueDate, ReturnDate

ERD



Referential Integrity

The use of foreign keys is applied to ensure that the Loan records reference valid members, copies and staff, and that each BookCopy references a valid BookTitle.

This is vital to prevent invalid references and to maintain database consistency.

Database Implementation

The database is named KhayelitshaLibraryDB and contains the required tables and sample records. The completed database includes at least five members, five book titles, eight copies, three staff members and five loans.

SQL Script

```
CREATE DATABASE KhayelitshaLibraryDB;
GO
USE KhayelitshaLibraryDB;
GO

CREATE TABLE Member (
  MemberID INT IDENTITY(1,1) PRIMARY KEY,
  FullName NVARCHAR(100) NOT NULL,
  Address NVARCHAR(250),
  Phone NVARCHAR(20),
  JoinDate DATE NOT NULL
);

CREATE TABLE BookTitle (
  BookTitleID INT IDENTITY(1,1) PRIMARY KEY,
  Title NVARCHAR(200) NOT NULL,
  Author NVARCHAR(150) NOT NULL,
  ISBN NVARCHAR(30),
  PublishedYear INT
);

CREATE TABLE BookCopy (
  CopyID INT IDENTITY(1,1) PRIMARY KEY,
  BookTitleID INT NOT NULL,
  Status NVARCHAR(20) NOT NULL,
  FOREIGN KEY (BookTitleID) REFERENCES BookTitle(BookTitleID)
);

CREATE TABLE Staff (
  StaffID INT IDENTITY(1,1) PRIMARY KEY,
  FullName NVARCHAR(100) NOT NULL
);

CREATE TABLE Loan (
  LoanID INT IDENTITY(1,1) PRIMARY KEY,
  MemberID INT NOT NULL,
  CopyID INT NOT NULL,
  StaffID INT NOT NULL,
  LoanDate DATE NOT NULL,
  DueDate DATE NOT NULL,
  ReturnDate DATE NULL,
  FOREIGN KEY (MemberID) REFERENCES Member(MemberID),
  FOREIGN KEY (CopyID) REFERENCES BookCopy(CopyID),
  FOREIGN KEY (StaffID) REFERENCES Staff(StaffID)
);
GO
```

Application Functionality

The dashboard provides navigation to Member Management, Book Management, Loan Management and Reports.

Member Management

The Member Form supports adding, reading, updating and deleting members. Members are searched by their name or Member ID. Validation prevents invalid phone numbers and incomplete input. Validation is applied to ensure that Members with loan history are protected from deletion.

Book Title and Copy Management

The Book Form manages book titles and physical copies separately. A title stores bibliographic information while each physical copy has its own status. The reason for it being applied through this method is to allow multiple copies of the same title to be managed separately.

Loan and Return Management

The Loan Form displays two tables one to display the available books and the other to display the current books that are being loaned. Issuing a book creates a loan and changes the copy status from Available to On Loan. Returning a book records the ReturnDate and changes the copy status back to Available. Historical loans are retained.

Search and Reports

The Reports Form provides member and book searches, loan filtering, overdue-book reporting and a loans per member summary. Loan records can be viewed as all, active, returned or overdue.

Database Connection and Security

The application uses Microsoft.Data.SqlClient and a connection string stored in App.config. Database operations use controlled SQL queries to ensure that the risk of SQL injection is handled. Connections, commands and transactions are managed safely.

Loan Transaction Logic

Issue and return operations use SQL transactions so related changes are committed together. Issuing verifies that a copy is Available, inserts the Loan record and changes the copy status to On Loan. Returning records ReturnDate and changes the copy status to Available. Failed operations can be rolled back.

Validation and Error Handling

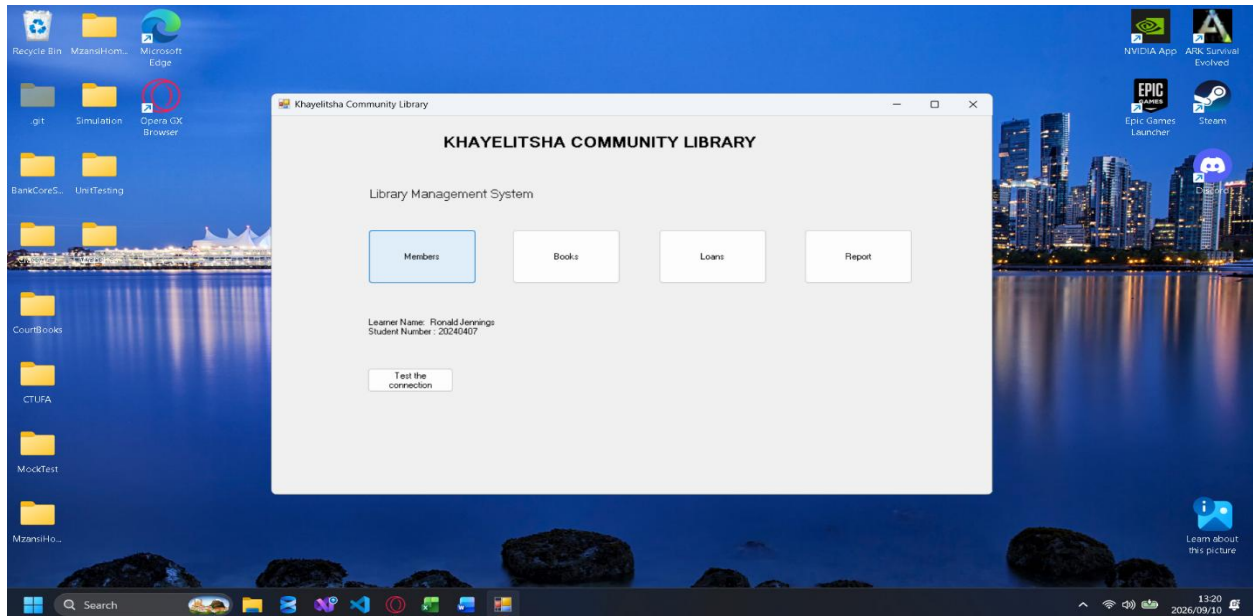
The required fields are checked, phone numbers are validated as 10 digits, and due dates are checked before a loan is issued. Operations in the application use exception handling and display user-friendly popup messages with clear descriptions of the error.

Design Decisions

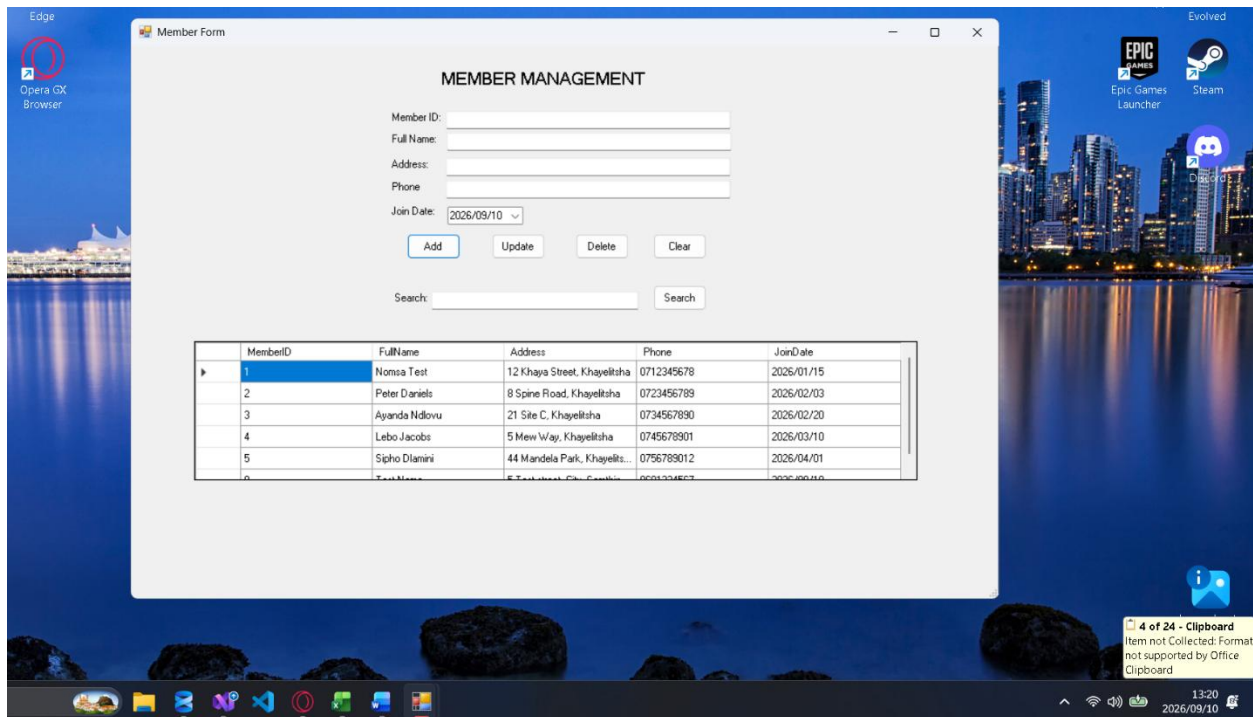
- The book title and physical book are kept separate to enable multiple copies.
- Available books and borrowed books are displayed separately.
- Historical borrowings are kept rather than being deleted.
- Transactions ensure consistency in Loan and BookCopy information.
- Controlled SQL ensures database manipulation entered by the user is safe.

Screenshots / Demonstration Evidence

MainForm



MemberForm



BookForm

Book Management

Book Title Information

Title:

Author:

ISBN:

Publication Year:

2026

Add

Update

Delete

Clear

Physical Book Copy Management

Copy ID:

Book Title:

Status:

Add Copy

Update

Delete

BookTitleID	Title	Author	ISBN	PublishedYear
1	To Kill a Mockingbird	Harper Lee	9780061120094	1960
2	1984	George Orwell	9780451524935	1949
3	The Alchemist	Paulo Coelho	9780062315007	1988
4	Long Walk to Freedom	Nelson Mandela	9780316548162	1994
5	The Hobbit	J.R.R. Tolkien	9780547928227	1937
6	The Great Gatsby	R. William Jennings	9780743273565	1925

CopyID	BookTitleID	Title	Status
1	1	To Kill a Mockingbird	On Loan
2	1	To Kill a Mockingbird	Available
3	2	1984	On Loan
4	2	1984	Available
5	3	The Alchemist	Available
6	4	Long Walk to Freedom	On Loan
7	5	The Hobbit	Available
8	5	The Hobbit	Available
9	1	To Kill a Mockingbird	Available

LoanForm

Loan and Return Management

Member:

Available Book:

Staff:

Loan Date:

2026/09/10

Due Date:

2026/09/24

Issue Book

Return Book

Clear

Available Books

Copy ID	Book Title	Author	Status
4	1984	George Orwell	Available
5	The Alchemist	Paulo Coelho	Available
7	The Hobbit	J.R.R. Tolkien	Available
8	The Hobbit	J.R.R. Tolkien	Available
2	To Kill a Mockingbird	Harper Lee	Available
9	To Kill a Mockingbird	Harper Lee	Available

Books Currently on Loan

Loan ID	Member	Copy ID	Book	Loan Date	Due Date
3	Ayanda Ndlovu	3	1984	2026/08/01	2026/08/15
4	Lebo Jacobs	6	Long Walk to Free...	2026/08/20	2026/09/03
6	Ayanda Ndlovu	1	To Kill a Mockingbird	2026/08/25	2026/09/08

ReportForm

Library Reports and Search

Member Search

Search

Clear

MemberID	FullName	Address	Phone	JoinDate
3	Ayanda Ndlovu	21 Site C, Khay...	0724567890	2026/02/20
4	Lebo Jacobs	5 Mew Way, Kh...	0745678901	2026/03/10
1	Nomsa Test	12 Khaya Street...	0712345678	2026/01/15
2	Peter Daniels	8 Spine Road, K...	0723456789	2026/02/03
5	Sipho Dlamini	44 Mandela Par...	0756789012	2026/04/01

Book Search

Search

Clear

BookTitleID	Title	Author	ISBN	PublishedYear
2	1984	George Orwell	9780451524325	1949
4	Long Walk to Free...	Nelson Mandela	9780316548182	1994
3	The Alchemist	Paulo Coelho	9780062315007	1988
6	The Great Gatsby	R. William Jennings	9780743273565	1925
5	The Hobbit	J.R.R. Tolkien	9780547328227	1937

Loan Report

Filter:

LoanID	Member	Book	CopyID	LoanDate	DueDate	ReturnDate	LoanStatus
1	Nomsa Test	To Kill a Mockingbird	1	2026/07/01	2026/07/15	2026/07/12	Returned
2	Peter Daniels	To Kill a Mockingbird	2	2026/07/10	2026/07/24	2026/07/22	Returned
8	Nomsa Test	To Kill a Mockingbird	2	2026/07/10	2026/07/24	2026/08/10	Returned
3	Ayanda Ndlovu	1984	3	2026/08/01	2026/08/15		Overdue
4	Lebo Jacobs	Long Walk to Freedom	6	2026/08/20	2026/09/03		Overdue

Overdue Books

Member	Book	CopyID	DueDate	DaysOverdue
Ayanda Ndlovu	1984	3	2026/08/15	26
Lebo Jacobs	Long Walk to Fre...	6	2026/09/03	7
Ayanda Ndlovu	To Kill a Mocking...	1	2026/09/08	2

Loans per Member

Member	TotalLoans	ActiveLoans	ReturnedLoans
Ayanda Ndlovu	4	2	2
Nomsa Test	2	0	2
Lebo Jacobs	1	1	0
Peter Daniels	1	0	1
Sipho Dlamini	1	0	1

13:21

2026/09/10

Conclusion

The Khayelitsha Community Library Management System has successfully implemented a relational SQL Server database and C# Windows Forms application. The system provides member, book, copy, loan and reporting functionality. CRUD operations, validation, parameterized SQL, exception handling and rules for the loan transaction were implemented. Testing has confirmed that the major functions are operational and operate correctly and that validation appropriately handles invalid inputs.

Testing Evidence

ID	Test	Input / Action	Expected / Actual Result	Status
TC01	Add Member – valid	Valid member details	Member inserted/displayed	PASS
TC02	Blank member name	Leave Full Name blank	Input rejected	PASS
TC03	Invalid phone	Non-10-digit phone	Validation message	PASS
TC04	Update Member	Change existing details	Record updated	PASS
TC05	Add Book Title	Valid title details	Title inserted/displayed	PASS
TC06	Add Book Copy	Select title and add copy	Copy created/displayed	PASS
TC07	Issue Book	Member + available copy + staff	Loan created; copy On Loan	PASS
TC08	Invalid Due Date	Invalid due date	Loan rejected	PASS
TC09	Return Book	Return active loan	ReturnDate saved; copy Available; current list refreshed	PASS
TC10	Search / Filter	Search/filter records	Matching records displayed	PASS